eptisa

FORM TECH-5 CURRICULUM VITAE FOR NOMINATED PERSONNEL

Page limit: Maximum of five (5) pages per CV (excluding part 14. Certification, which does not count toward the five-page limit)

NOTE TO THE CV EDITOR: PLEASE REVIEW THE CV FORMAT AGAINST THE REQUIREMENTS SPECIFIED IN THE RFP AND ENSURE THAT THE CERTIFICATIONS SECTION IS FULLY COMPLIANT AND ALIGNED WITH THE REQUESTED FORMAT AND CRITERIA. PLEASE VERIFY THIS CAREFULLY BEFORE FINALIZING THE CV.

PLEASE ENSURE THAT THIS NOTE AND ALL HIGHLIGHTED SECTIONS ARE REMOVED BEFORE ADDING THE CV TO THE FINAL DOSSIER.

1. **Proposed Position:** AI & DATA SCIENCE BACKEND DEVELOPER
2. **Name of Entity:** Eptisa
3. **Name of Expert:** Elena Avilés Gil
4. **Date of Birth:** **Citizenship:**
5. **Complete personal contact details:** , Seville Spain
   (+34) 665 999 455 | elenaavilesgil@gmail.com
6. **Education:**
   09-2020 – 06-2025 — Universidad de Sevilla; Double Degree in Computer Engineering (IT) & Mathematics
7. **Membership in Professional Associations:**
   Mindbreeze Training — Mindbreeze
8. **Other Training:**
   Python; Java; JavaScript; CSS; HTML; SQL; R; MATLAB; Teamwork; Analytical thinking & problem-solving; Fast learning and adaptability; Creativity
9. **Countries of Work Experience:**
10. **Languages:** Level of Proficiency (mother tongue, excellent, good, fair, poor)

Language	Speaking	Reading	Writing
Spanish	Excellent	Excellent	Excellent
English	Excellent	Excellent	Excellent

11. **Employment Record:**

1. **From [Month, Year]:** 02-2025 **To [Month, Year]:** 08-2025
   Employer: Eptisa
   Positions held: AI Intern – Web Development
2. **From [Month, Year]:** 08-2025 **To [Month, Year]:** Present
   Employer: Eptisa
   Positions held: Backend Developer specializing in AI & Data Science

12. Detailed Tasks Assigned	13. Work Undertaken that Best Illustrates Capability to Handle the Tasks Assigned

	Name of assignment or project: RAG System (Retrieval-Augmented Generation) **Year:** null **Location:** **Client:** **Employer:** Eptisa **Main project features:** Development of a RAG-based intelligent search system, combining LLMs with information retrieval techniques to improve response accuracy and relevance. Implementation of data ingestion, segmentation, and indexing pipelines, transforming unstructured data into vector representations for efficient querying. Design of a semantic search system using embeddings, integrating context retrieval into the response generation process and optimizing coherence and accuracy across different document types. **Positions held:** Backend Developer specializing in AI & Data Science **Activities performed:** Development of a RAG-based intelligent search system; Implementation of data ingestion, segmentation, and indexing pipelines; Design of a semantic search system using embeddings
Name of assignment or project: Traffic Sign Detection and Classification (YOLO) **Year:** null **Location:** **Client:** **Employer:** Eptisa **Main project features:** Development of a computer vision system for traffic sign detection and classification using Deep Learning models such as YOLO. Data preparation and preprocessing, along with model training and validation to improve detection performance. Implementation of post-processing logic to extract relevant information and enhance accuracy, integrating the model into a complete image processing pipeline from input data to final outputs. **Positions held:** Backend Developer specializing in AI & Data Science **Activities performed:** Development of a computer vision system for traffic sign detection and classification; Data preparation and preprocessing, model training and validation; Implementation of post-processing logic	
Name of assignment or project: EMT Madrid Data Analysis **Year:** null **Location:** **Client:** **Employer:** Eptisa **Main project features:** Development of a data analysis project applied to public transportation, focused on identifying patterns and supporting data driven decision-making. Exploration, cleaning, and transformation of EMT datasets, along with the implementation of processes for handling large volumes of data. Application of dimensionality reduction techniques (PCA) and clustering methods for data segmentation, combined with data analysis and visualization tools to extract meaningful insights. **Positions held:** Backend Developer specializing in AI & Data Science **Activities performed:** Development of a data analysis project applied to public transportation; Exploration, cleaning, and transformation of EMT datasets; Implementation of processes for handling large volumes of data; Application of dimensionality reduction techniques (PCA) and clustering methods; Data analysis and visualization tools to extract meaningful insights	

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Name of assignment or project: Pavement Algorithms Migration (VBA → Python)
Year: null
Location:
Client:
Employer: Eptisa
Main project features: Migration and optimization of a road deterioration simulation model, translating complex logic from VBA to Python while improving performance and scalability. Refactoring of the system to enhance readability, maintainability, and efficiency, while preserving consistency with the original model. Implementation of calculation functions, yearly simulation, and parameter adjustment, including optimization processes using numerical methods and efficient data structures for handling large datasets.
Positions held: Backend Developer specializing in AI & Data Science
Activities performed: Migration and optimization of a road deterioration simulation model; Refactoring of the system to enhance readability, maintainability, and efficiency; Implementation of calculation functions, yearly simulation, and parameter adjustment; Optimization processes using numerical methods and efficient data structures

Name of assignment or project: Bachelor's Thesis – Voronoi Diagrams
Year: null
Location:
Client: Andalusian Center for Developmental Biology (CABD)
Employer: Eptisa
Main project features: Development of a project focused on Voronoi diagrams, combining computational geometry and Artificial Intelligence techniques for spatial structure segmentation and analysis. Implementation of geometric algorithms for the construction and analysis of Voronoi diagrams, applied to spatial point distribution and partition optimization. Integration of Deep Learning models such as StarDist for object detection and segmentation, combining algorithmic and machine learning approaches with a focus on computational efficiency and accuracy. Collaboration in a multidisciplinary environment, adapting technical solutions to different requirements.
Positions held: Backend Developer specializing in AI & Data Science
Activities performed: Development of a project focused on Voronoi diagrams; Implementation of geometric algorithms for the construction and analysis of Voronoi diagrams; Integration of Deep Learning models such as StarDist for object detection and segmentation; Collaboration in a multidisciplinary environment

14.	**Certification:**

I, the undersigned, certify to the best of my knowledge and belief:

i. this CV correctly describes my qualifications and my experience.
ii. I am not a current or former public sector employee of a government department or agency of the Beneficiary (including the Executing or Implementing Agency).
   - If yes, declare relevant details including agency, role, and employment period: _____
iii. I am not a close relative (other than spouse) of a current ADB staff member.
iv. I am not a spouse of a current ADB staff member.
   - If yes, declare the staff member: _____
v. I am not a former ADB staff member or executive director or alternate executive director of its Board of Directors.
   - If yes, I separated from ADB over 12 months ago: _____
vi. I was not directly or indirectly involved in any part of the preparation of the terms of reference for this consulting services assignment.
vii. I will not be in a situation of actual, potential, or perceived conflict of interest by undertaking this assignment.
   - If yes or unsure, please disclose details of the conflict: _____
viii. I am not debarred or temporarily suspended by ADB (ineligible for ADB engagement).
ix. I am not listed in the United Nations Security Council Consolidated List and I am not from a country which is prohibited to export goods to or receive any payments from the Beneficiary's

country as an act of compliance with a decision of the United Nations Security Council taken under Chapter VII of the Charter of the United Nations.

x. I am not, nor have I ever been, temporarily suspended, debarred, declared ineligible, or blacklisted by any country, any international organization, and/or any other development or donor agency.
   - If so debarred, declared ineligible, temporarily suspended, or blacklisted, please state details (as applicable to each):
     - Name of institution: _____
     - Period of debarment, ineligibility, or blacklisting (start and end date): ____
     - Reason for debarment, ineligibility, or blacklisting: _____

xi. I have not been charged or convicted of any criminal offense (including misdemeanors) or infractions or violations of ordinance that carry the penalty of imprisonment.
   - If so charged or convicted, please state details (as applicable to each):
     - Nature of the offense or violation: _____
     - Court or area of jurisdiction: _____
     - Resolution (i.e., dismissed; settled; convicted, duration of penalty): _____
     - Other relevant details: _____

[If CV is signed by the individual:]

I certify that I have been informed by the entity that is including my CV in the Proposal for the [*name of the project*]. I confirm that I will be available to carry out the assignment for which my CV has been submitted, in accordance with the implementation arrangements and schedule set out in the Proposal.

[If CV is signed by the Consultant's authorized representative and the written agreement attached:]

I, as the authorized representative of the Consultant submitting this Proposal for the [*title of the assignment*] certify that I have obtained the consent of the named expert to submit his/her CV, and that I have obtained a written representation from the expert that s/he will be available to carry out the assignment in accordance with the implementation arrangements and schedule set out in the Proposal.

I understand that any misrepresentation that knowingly or recklessly misleads, or attempts to mislead, may lead to disqualification or dismissal, and/or the imposition of sanctions or other remedial actions by ADB.

Name of Expert: Elena Avilés Gil	*Day/Month/Year*
Sign:	
Full name of authorized representative:	*Day/Month/Year*
Sign:	